Neologik Onboarding - Quick Reference

Overview

This script creates resources and assigns permissions for Neologik deployment in a customer's Azure tenant.

Run: `.\Install-NeologikEnvironment.ps1` in PowerShell 7 as Administrator

Prerequisites

Required Permissions to Run Script

User must have:

- **Owner** role at subscription level
- **Global Administrator** role in Entra ID

Reason: Script assigns Azure RBAC roles (requires Owner) and Microsoft Graph app-role assignments plus enterprise-application group assignments (require Global Administrator)

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vCPU Quota

Neologik requires sufficient vCPU quota in the target subscription and region before deployment can proceed. Check and request increases **before** running the onboarding script.

Checking current quota

1. Azure Portal → **Subscriptions** → select your subscription → **Usage + quotas**
2. Filter by your target region and search for `Standard D2s v6` (the Dsv6 family) to see current usage vs. limit

Requesting an increase

Azure Portal → **Usage + quotas** → select the quota line → **New Quota Request**

Variables Used

- `{org}` = Organization code (3 characters, lowercase)
- `{region}` = Azure region abbreviation (e.g., `uks`, `eus`, lowercase)
- `{index}` = Environment index (e.g., `01`, `02`)
- `{env}` = Environment type (`dev`, `prd`)
- `{short}` = Solution short name: `{org}{env}{index}` (e.g., `abcprd01`)

The environment type joins the organisation code in every name (project#1576). Every computed name below - including the resource group - can be overridden with user input to match a client-mandated

naming convention.

Example: Organization "ABC", UK South, index 01

- Resource Group: `rg-neo-abcprd-uks-01`
- Key Vault: `kvabcprd01`
- Storage Account: `stneodeployabcprduks01`
- Default User Group: `Neologik User Group - abcprd01`

Changes Applied by Script

1. Guest User Invitations (Entra ID) - optional

Action: Invite Neologik users as B2B guests (if no account with that email already exists)

Source: Email addresses loaded from `NeologikGuestUsers.txt` file

Notes:

- Accounts you have pre-created (B2B guests or full member accounts whose mail or UPN matches a listed address) are detected and used as-is - no duplicate invitation
- Interactive runs list the file's addresses and offer to **add more to invite**, typed one per line (blank line finishes, commas accepted within a line); with an empty or missing file the whole invite list can be typed in
- Skip invitations entirely (user input) when you provision accounts in your own tenant; groups are then seeded with the signed-in user only
- Declining invitations offers a follow-up: **add existing pre-created accounts to the security groups** with no invitation sent - addresses from `NeologikGuestUsers.txt`. Addresses with no matching account are reported and skipped; the run continues
- All accounts processed by this step (invited or pre-existing) join **all four groups**, including the Admin User Group and its Azure RBAC - Neologik admin/support accounts only

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2. Security Groups (Entra ID)

Action: Create four security groups

Group Name	Purpose	Members
`Neologik User Group - {short}`	Default users group - assigned to agent channels created post-deployment in the Neologik admin tool (NCE). The platform never creates groups; channels reuse this group or another existing group you supply	Current user + Neologik users
`Neologik NCE Contributors - {short}`	NCE contributor role (create and change platform	Current user + Neologik users

Group Name	Purpose	Members
	configuration). Renamed in place from the pre-v5 `Neologik NCE User Group` where one exists	
`Neologik NCE Admins - {short}`	NCE admin role (Blueprints, Audit, Admin page)	Current user + Neologik users
`Neologik Admin User Group - {short}`	Azure resource administration	Current user + Neologik users

Note: No per-bot groups are created - agents and their channels are created after deployment in the NCE.

Scope: Entra ID tenant level

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3. Resource Group

Action: Create resource group

Name	Location
`rg-neo-{org}{env}-{region}-{index}` (name can be overridden with user input)	Customer specified (e.g., uksouth)

Scope: Subscription

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4. Role Assignments - Security Groups

Action: Assign roles to Neologik Admin User Group (all at **resource group** scope; no subscription-scoped roles)

Role	Scope	Principal
Contributor	Resource Group	Neologik Admin User Group
Azure Kubernetes Service Cluster Admin Role	Resource Group	Neologik Admin User Group
Azure Kubernetes Service RBAC Cluster Admin	Resource Group	Neologik Admin User Group
Key Vault Administrator	Resource Group	Neologik Admin User Group
Storage Blob Data Owner	Resource Group	Neologik Admin User Group
Storage Table Data Contributor	Resource Group	Neologik Admin User Group

Role	Scope	Principal
Cognitive Services Data Reader	Resource Group	Neologik Admin User Group
Cognitive Services OpenAI Contributor	Resource Group	Neologik Admin User Group
Search Index Data Reader	Resource Group	Neologik Admin User Group
Azure AI Developer	Resource Group	Neologik Admin User Group

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5. App Registration & Service Principal (Entra ID)

Action: Create app registration for GitHub deployments - or reuse a service principal you have already created, supplied as user input (creation steps in CREATE-SP-GUIDE.md)

Property	Value
Name	`Neologik GitHub Service Connection - {short}`
Type	Multi-tenant (AzureADMultipleOrgs)
Client Secret	Created (365-day expiration); with a client-provided SP the secret is optional and client-managed
Secret Storage	Stored in Key Vault (`neologik-deployment-service-principle-secret`)

Group Membership

- Added to: `Neologik Admin User Group - {short}`

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6. Role Assignments - Service Principal

Action: Assign roles to the deployment service principal

Role	Scope	Principal
Reader	Subscription	Service Principal
Cost Management Contributor	Subscription	Service Principal
Contributor	Resource Group	Service Principal
Key Vault Certificates Officer	Resource Group	Service Principal
Key Vault Secrets Officer	Resource Group	Service Principal

Role	Scope	Principal
Storage Blob Data Contributor	Resource Group	Service Principal
Azure Kubernetes Service RBAC Cluster Admin	Resource Group	Service Principal
Monitoring Contributor	Resource Group	Service Principal
Role Based Access Control Administrator	Resource Group	Service Principal
Application.Read.All	Microsoft Graph (app role)	Service Principal
Application.ReadWrite.OwnedBy	Microsoft Graph (app role)	Service Principal
AgentIdentityBlueprint.Create	Microsoft Graph (app role)	Service Principal
AgentIdentityBlueprint.AddRemoveCreds.All	Microsoft Graph (app role)	Service Principal
AgentIdentityBlueprint.UpdateAuthProperties.All	Microsoft Graph (app role)	Service Principal
AgentIdentityBlueprintPrincipal.Create	Microsoft Graph (app role)	Service Principal
AgentIdentity.Create.All	Microsoft Graph (app role)	Service Principal
AgentIdUser.ReadWrite.All	Microsoft Graph (app role)	Service Principal

> The subscription-scoped roles are read-only (**Reader**, for SKU/quota pre-flight checks) and cost-configuration-only (**Cost Management Contributor**, for the scheduled cost-report emails) - all resource writes are resource-group-scoped. The six **AgentIdentity*/AgentIdUser** roles let the deployment service principal provision **Microsoft Entra Agent ID** blueprints, principals, and agent users for autonomous agents - scoped to agent-id only, rather than granting the broad "Agent ID Administrator" directory role. No group-management permission is granted: the platform never creates Entra groups.

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7. Platform Sign-in App Registration (Entra ID)

Action: Create the sign-in app used by the Neologik admin tool (NCE) and App portal login

Property	Value
Name	`Neologik Platform Sign-in - {short}`
Type	Multi-tenant (AzureADMultipleOrgs), web redirect URI on your hostname

Property	Value
Client Secret	Created (365-day expiration), stored in Key Vault
Owner	Deployment service principal, on both the application and its service principal objects (so the pipeline can manage/rotate it)
Group Assignments	`Neologik NCE Contributors - {short}` and `Neologik NCE Admins - {short}` assigned to the enterprise application, so members' tokens carry the groups claim NCE authorizes from

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8. Key Vault

Action: Create Key Vault with RBAC authorization

Property	Value
Name	`kv{org}{env}{index}` (name can be overridden with user input)
Authorization	Microsoft Entra ID RBAC (no access policies)
Public Access	Enabled
Soft Delete	Enabled (Azure default)

Secrets Stored

Secret Name	Content
`neologik-deployment-service-principle-secret`	Deployment service principal client secret
`sp-admin-{solution-name}-auth-secret`	Platform Sign-in app client secret

Role Assignments

Role	Scope	Principal
Key Vault Secrets Officer	Resource Group	Neologik Admin User Group

Note: Allows group members to read/write secrets in Key Vault

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9. Storage Account

Action: Create storage account for the certificate and configuration handoff

Property	Value
Name	`stneodeploy{org}{env}{region}{index}` (truncated to 24 characters; name can be overridden with user input)
Authentication	Microsoft Entra ID only
Shared Key Access	Disabled
Public Access	Enabled
Blob Containers	`certificate` (TLS certificate), `json` (configuration JSON handoff), `logs`

Role Assignments

Role	Scope	Principal
Storage Blob Data Contributor	Resource Group	Neologik Admin User Group

Note: Allows group members to upload/manage certificates

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10. Managed Identities (User-Assigned)

Action: Create 2 managed identities

Identity 1: Script Runner Service Connection

Property	Value
Name	`neologik-script-runner-service-connection-{short}` (name can be overridden with user input)
Display Name	`Neologik Script Runner Service Connection - {short}`

Role Assignments:

Role	Scope
Contributor	Subscription
Key Vault Secrets Officer	Resource Group
Azure Bot Service Contributor Role	Resource Group

Identity 2: SQL Managed Identity

Property	Value
Name	`neologik-sql-managed-identity-{short}` (name can be overridden with user input)
Display Name	`Neologik SQL Managed Identity - {short}`

Role Assignments:

Role	Scope
Directory Readers	Entra ID (Directory Role)

11. Resource Provider Registration

The script registers a minimal set of Azure resource providers on the subscription (Compute, KeyVault, ManagedIdentity, Resources, Storage), or the full platform set of around 30 if the operator opts in at the prompt. Registration enables resource types on the subscription; it grants no permissions and creates no resources.

Summary of Permissions by Scope

Subscription Level

Principal	Role
Service Principal (GitHub)	Reader
Service Principal (GitHub)	Cost Management Contributor
Managed Identity (Script Runner)	Contributor

Resource Group Level

Principal	Role
Service Principal (GitHub)	Contributor
Service Principal (GitHub)	Key Vault Certificates Officer
Service Principal (GitHub)	Key Vault Secrets Officer
Service Principal (GitHub)	Storage Blob Data Contributor
Service Principal (GitHub)	Azure Kubernetes Service RBAC Cluster Admin
Service Principal (GitHub)	Monitoring Contributor
Service Principal (GitHub)	Role Based Access Control Administrator
Managed Identity (Script Runner)	Key Vault Secrets Officer

Principal	Role
Managed Identity (Script Runner)	Azure Bot Service Contributor Role
Neologik Admin User Group	Contributor + the administration role set in section 4

Microsoft Graph (App Roles)

Principal	Role
Service Principal (GitHub)	Application.Read.All
Service Principal (GitHub)	Application.ReadWrite.OwnedBy
Service Principal (GitHub)	AgentIdentityBlueprint.Create
Service Principal (GitHub)	AgentIdentityBlueprint.AddRemoveCreds.All
Service Principal (GitHub)	AgentIdentityBlueprint.UpdateAuthProperties.All
Service Principal (GitHub)	AgentIdentityBlueprintPrincipal.Create
Service Principal (GitHub)	AgentIdentity.Create.All
Service Principal (GitHub)	AgentIdUser.ReadWrite.All

Entra ID (Directory Roles)

Principal	Role
Managed Identity (SQL)	Directory Readers